Calabi–Yau quantum groups and 6d holomorphic Chern–Simons

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Abstract

Volume I of the modular Koszul duality programme introduces E_1-chiral quantum groups as abstract algebraic-geometric objects: ordered bar complexes with spectral R-matrices, Drinfeld coproducts, and Yang–Baxter equations on configuration spaces of curves. This paper constructs them concretely from Calabi–Yau geometry.

The source is the holomorphic Chern–Simons hierarchy. At the top ($d = 6$): holomorphic Chern–Simons theory on a CY_3-fold Y. A curve $C \subset Y$ of codimension 2 carries a defect algebra: the normal bundle $N_{C/Y}$ supplies two spectral parameters, and the defect algebra is a chiral algebra on C with quantum group structure from the normal directions. For $Y = \mathbb{C}^3$ and $C = \mathbb{C}$, the defect algebra is $\mathcal{W}_{1+\infty}$ at $\Psi = -\sigma_2(h_1, h_2, h_3)$. The hierarchy $6 \to 5 \to 4 \to 3$ is codimension-2 restriction at each step; each restriction produces a chiral algebra from a gauge theory, and the E_n level decreases as the dimension drops.

The critical cohomological Hall algebra $\mathrm{CoHA}(Q, W)$ of a quiver with CY_3 potential is the E_1-primitive: its product is ordered Hall multiplication, its R-matrix comes from the Ext-quiver, and its coproduct is the Schiffmann–Vasserot vertex coproduct. For the Jordan quiver: $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The Drinfeld center recovers E_2-braiding, closing the E_n operadic circle.

Quantum toroidal algebras $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_1)$ carry the CY condition as the parameter constraint $q_1 q_2 q_3 = 1$. The structure function $G(x; q_1, q_2, q_3) = \prod_i (1 - q_i x) / \prod_i (1 - q_i^{-1} x)$ satisfies the CY reflection identity $G(x) \cdot G(1/x) = 1$ precisely when $q_1 q_2 q_3 = 1$.

Wall-crossing in the Bridgeland stability manifold is Maurer–Cartan gauge equivalence in the convolution algebra of Volume I: the Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation $\Theta_\sigma \sim \Theta_{\sigma'}$. The shadow obstruction tower at $\kappa_{\mathrm{ch}} = \Psi$ produces the perturbative constant-map Gromov–Witten free energies $F_g^{\mathrm{GW,const}}(\mathbb{C}^3)$: for $\mathbb{C}^3$ (no compact curves), the shadow IS the full GW answer.

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1 Introduction

1.1 CY quantum groups as concrete E_1-chiral quantum groups

Volume I constructs E_1-chiral quantum groups as abstract algebraic-geometric objects on curves. An E_1-chiral quantum group is a tuple $(A, \mu, \Delta_z, \varepsilon, \eta, S)$ satisfying five axioms (H1–H5): an E_1-chiral algebra structure, a spectral coproduct, bialgebra compatibility, spectral coassociativity

$$(\Delta_w \otimes \mathrm{id}) \circ \Delta_{z+w} = (\mathrm{id} \otimes \Delta_z) \circ \Delta_w,$$

and the Hopf axiom at $z = 0$. The ordered bar $B^{\mathrm{ord}}(A) = T^c(s^{-1}\bar{A})$ preserves the R-matrix; the symmetric bar $B^{\Sigma}(A)$ kills the Hopf structure via the lossy averaging map: in abelian and scalar families $\mathrm{av}(r(z)) = \kappa_{\mathrm{ch}}$, while for non-abelian affine Kac–Moody $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa_{\mathrm{ch}}$.

Neither Volume I nor Volume II constructs a single E_1-chiral quantum group. This paper constructs them from two sources: the 6d holomorphic Chern–Simons hierarchy (which produces chiral algebras as defect algebras on codimension-2 curves) and the critical cohomological

Hall algebra (which provides the E_1-primitive as an associative algebra graded by dimension vectors).

The E_1 stabilization theorem of the companion paper [**?**] proves that for $d \geq 3$, the CY chiral algebra is natively E_1: the Schouten–Nijenhuis bracket vanishes and all noncommutative structure arises from quantization. The braided E_2-structure is recovered through the Drinfeld center of the E_1-monoidal representation category, not through the native E_d-framing (which produces only symmetric braiding for $d \geq 3$, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$).

1.2 The holomorphic Chern–Simons hierarchy

The theories of this paper form a hierarchy in real dimension:

d	Theory	Curve output	E_n level
6	hCS on CY_3 Y	defect on $C \subset Y$	E_1
5	partially topological CS on $\mathbb{R} \times S$	$\mathbb{R} \times C$	E_1
4	hCS on $\mathbb{C}^2$ (CWY)	line operator	E_1
3	3d HT (Volume II)	boundary algebra	E_1

The hierarchy $6 \to 5 \to 4 \to 3$ is codimension-2 restriction at each step. Each restriction produces a chiral algebra from a gauge theory. The resulting chiral algebras are E_1-chiral quantum groups in the sense of Volume I, with R-matrices from the normal directions and coproducts from the spectral parameters.

1.3 The E_n operadic circle

The E_n hierarchy organizes the entire programme:

$$E_3(\text{bulk}) \;\to\; E_2(\text{boundary chiral}) \;\to\; E_1(\text{bar/QG}) \;\to\; E_2(\text{Drinfeld center}) \;\to\; E_3(\text{derived center}).$$

Each arrow has precise mathematical content: $E_3 \to E_2$ is restriction to a codimension-2 defect; $E_2 \to E_1$ is the ordered bar complex; $E_1 \to E_2$ is the Drinfeld center; $E_2 \to E_3$ is the higher Deligne conjecture. The circle closes for 3d HT theories with conformal vector at non-critical level (the topologization theorem, proved for affine KM); without conformal vector, the theory is stuck at $\mathrm{SC}^{\mathrm{ch,top}}$.

The CY quantum groups of this paper enter the circle at the E_1 vertex: the CoHA is the E_1-primitive, and the Drinfeld center carries it to E_2. The tetrahedron equation (3d analogue of YBE) fails for the Yang R-matrix at $O(\kappa_{\mathrm{ch}}^2)$: the E_3 structure is genuinely nontrivial beyond E_2.

1.4 Outline

Section 2 develops the hCS hierarchy from 6d to 3d with the codimension-2 defect mechanism. Section 3 constructs the codimension-2 defect algebras and proves the $\mathcal{W}_{1+\infty}$ identification for $\mathbb{C}^3$. Section 4 establishes the CoHA as the E_1-primitive and develops the Hopf axioms. Section 5 treats quantum toroidal algebras and the CY condition $q_1 q_2 q_3 = 1$. Sections 6–7 develop the Drinfeld center and the shadow-equals-GW identification.

2 The holomorphic Chern–Simons hierarchy

2.1 6d holomorphic Chern–Simons

Let Y be a CY_3-fold with holomorphic volume form $\Omega_Y \in H^0(Y, \omega_Y)$. The holomorphic Chern–Simons functional on Y (Witten, Costello) is

$$S_{\mathrm{hCS}}(\mathcal{A}) = \int_Y \Omega_Y \wedge \mathrm{tr}\Big(\mathcal{A} \wedge \bar{\partial}\mathcal{A} + \tfrac{2}{3}\mathcal{A} \wedge \mathcal{A} \wedge \mathcal{A}\Big)$$

where $\mathcal{A} \in \Omega^{0,1}(Y, \mathfrak{g})$ is a $(0,1)$-form valued in a Lie algebra $\mathfrak{g}$. The equations of motion $\bar{\partial}\mathcal{A} + \mathcal{A} \wedge \mathcal{A} = 0$ are the integrability condition for the holomorphic structure on the trivial G-bundle.

The theory is six-dimensional, holomorphic in all three complex directions. The BRST operator is $Q_{\mathrm{BRST}} = \bar{\partial} + [\mathcal{A}, -]$, and the observables are the Dolbeault cohomology $H^{0,\bullet}_{\bar{\partial}_{\mathcal{A}}}(Y, \mathfrak{g})$.

Remark 2.1 (Relation to physical dimensions). The 6d hCS theory is not a physical quantum field theory in the Wightman sense: the kinetic term is first-order, the path integral is one-loop exact, and the partition function is a holomorphic invariant. It is the holomorphic twist of the 6d $(2,0)$ theory on a CY_3-fold, and its observables produce chiral algebras by restriction to codimension-2 defects. The physical content is in the defect algebras, not in the bulk theory.

2.2 Dimensional reduction: $6 \to 5 \to 4 \to 3$

The hierarchy proceeds by codimension-2 restriction at each step.

$6 \to 5$: for a CY_3-fold of the form $Y = \mathbb{R} \times S$ (where S is a CY_2 surface), the 6d hCS theory reduces to 5d partially topological Chern–Simons on $\mathbb{R} \times S$. The topological direction $\mathbb{R}$ carries the E_1 ordering; the holomorphic surface S carries the chiral data. Restriction to $\mathbb{R} \times C$ (where $C \subset S$ is a curve) produces the ordered bar complex with one spectral parameter.

$5 \to 4$: for $S = \mathbb{C}^2$, the 5d theory reduces to 4d holomorphic Chern–Simons on $\mathbb{C}^2$: the Costello–Witten–Yamazaki theory. Line operators on $\mathbb{C} \subset \mathbb{C}^2$ produce Yangians from holomorphic gauge fields. The R-matrix comes from the normal direction to the line.

$4 \to 3$: restricting to a further codimension-2 locus produces the 3d holomorphic-topological theories of Volume II. The boundary algebra is a chiral algebra on the codimension-2 curve, and the $\mathrm{SC}^{\mathrm{ch,top}}$ pair $(C^\bullet_{\mathrm{ch}}(A, A), A)$ is the boundary–bulk system.

Proposition 2.2 (Dimensional consistency)**.** *At each step of the hierarchy, the E_n level of the chiral algebra on the defect curve matches the ambient dimension: 6d produces E_1-chiral (two normal directions give two spectral parameters, but the CY condition reduces to E_1); 5d produces E_1-chiral with one spectral parameter; 4d produces E_1-chiral (Yangian); 3d produces E_1-chiral (boundary algebra of Volume II). The E_2-braiding is recovered at each level through the Drinfeld center.*

2.3 The Ω-background and equivariant parameters

For the toric CY_3-fold $Y = \mathbb{C}^3$, the three coordinate directions carry equivariant parameters $h_1, h_2, h_3 \in \mathbb{C}$ subject to the CY condition $h_1 + h_2 + h_3 = 0$. In the multiplicative parametrization $q_i = e^{h_i}$, the CY condition reads $q_1 q_2 q_3 = 1$. The identification $q_1 = q$, $q_2 = t^{-1}$, $q_3 = t/q$ recovers the standard (q,t)-parametrization of the quantum toroidal algebra.

The Ω-background parameters enter the defect algebra through the equivariant Chern–Simons functional: the gauge field on Y decomposes along the normal bundle to a defect curve $C \subset Y$, and the equivariant parameters weight the normal modes. For $C = \mathbb{C}_z$ (the first coordinate axis) inside $Y = \mathbb{C}^3$, the normal bundle is $N_{C/Y} = \mathcal{O}_C^{\oplus 2}$ with fiber coordinates (w_1, w_2) carrying equivariant weights (h_2, h_3). The resulting defect algebra on C has OPE poles involving both normal parameters.

3 Codimension-2 defect algebras

3.1 The defect construction

A curve $C \subset Y$ in a CY_3-fold carries a defect algebra: the 6d hCS theory restricted to a tubular neighborhood of C produces a 2d chiral algebra on C with additional structure from the normal bundle.

Construction 3.1 (Codimension-2 defect algebra)**.** Let Y be a CY_3-fold, $C \subset Y$ a smooth curve, and $N_{C/Y}$ the normal bundle of rank 2. The defect algebra A_C^{def} on C is obtained by:

Step 1. Restriction. Restrict the 6d hCS gauge field $\mathcal{A} \in \Omega^{0,1}(Y, \mathfrak{g})$ to the formal neighborhood $\hat{N}_{C/Y}$ of C in Y. The Taylor expansion along the normal fibers gives a collection of fields on C indexed by normal-direction monomials.

Step 2. BRST reduction. The BRST operator $Q = \bar{\partial} + [\mathcal{A}, -]$ restricts to the formal neighborhood. The Q-cohomology in the fiber directions produces the generating fields of the defect algebra on C.

Step 3. OPE from normal modes. The OPE on C is inherited from the 6d hCS action. The normal bundle $N_{C/Y}$ supplies two spectral parameters (from the two fiber coordinates), which enter the OPE as meromorphic functions of the insertion points on C.

Remark 3.2 (Two spectral parameters)*.* The rank-2 normal bundle $N_{C/Y}$ gives two spectral parameters. This is the geometric origin of the two-parameter family of quantum toroidal algebras: the two normal directions to a curve inside a CY_3-fold produce the (q, t)-deformation. For a CY_2-fold (surface), the normal bundle has rank 1, producing one spectral parameter: the standard spectral parameter of Yangians and KZ equations.

3.2 $\mathbb{C}^3$: the $\mathcal{W}_{1+\infty}$ identification

Proposition 3.3 (Defect algebra of $\mathbb{C}^3$)**.** *For $Y = \mathbb{C}^3$ with coordinates (z, w_1, w_2) and CY form $\Omega = dz \wedge dw_1 \wedge dw_2$, the codimension-2 defect algebra on $C = \mathbb{C}_z$ (the z-axis) is the $\mathcal{W}$-algebra $\mathcal{W}_{1+\infty}$ at deformation parameter*

$$\Psi = -\sigma_2(h_1, h_2, h_3)$$

where σ_2 is the second elementary symmetric polynomial in the Ω-background parameters. The central charge is $c_{\mathcal{W}} = 1$ (the Sugawara value from the $\mathfrak{gl}_1$ current subalgebra).

Proof sketch. The gauge field $\mathcal{A} \in \Omega^{0,1}(\mathbb{C}^3, \mathfrak{gl}_N)$ expands along the normal fiber as $\mathcal{A}(z, w_1, w_2) = \sum_{a,b \geq 0} J_{a,b}(z)\, w_1^a w_2^b\, d\bar{z}$, where $J_{a,b}$ are fields on $C = \mathbb{C}_z$. The BRST cohomology in the (w_1, w_2)-directions retains one field at each bi-degree (a, b) with $a + b = s - 1$ for spin s: these are the

generators of $\mathcal{W}_{1+\infty}$. At spin 1: a single current $J(z)$ (the $\mathfrak{gl}_1$ current). At spin 2: the stress tensor $T(z)$ and the $U(1)$ current. The OPE is computed from the 6d hCS action by restricting to the formal neighborhood; the resulting algebra is $\mathcal{W}_{1+\infty}$ at $\Psi = -\sigma_2 = -(h_1h_2+h_1h_3+h_2h_3)$.

The CY condition $h_1+h_2+h_3 = 0$ gives $\sigma_2 = -(h_1^2+h_2^2+h_3^2)/2$, so $\Psi = (h_1^2+h_2^2+h_3^2)/2 > 0$ generically. The central charge of the $\mathcal{W}_{1+\infty}$ algebra at $N = 1$ (single $\mathfrak{gl}_1$ current) is computed from the Sugawara construction $T = (1/(2\Psi)) : J^2 :$ applied to the level-Ψ Heisenberg current $J(z)J(w) \sim \Psi/(z-w)^2$; this gives $c = 1$ for every nonzero Ψ, independent of the specific value. For N bosons the central charge is $c = N$; the 6d $\mathfrak{gl}_1$ gauge reduction produces $N = 1$. □

Remark 3.4 (Level-prefixed r-matrix). The classical r-matrix of the $\widehat{\mathfrak{gl}}_1$ current subalgebra derived from the 6d reduction carries level $\Psi = -\sigma_2$, not a normalised level $k = 1$:

$$r^{\mathrm{Heis}}(z) = \frac{\Psi}{z} = \frac{-\sigma_2(h_1, h_2, h_3)}{z}.$$

At $\Psi = 0$ the r-matrix vanishes and the Sugawara construction is singular; this is the degenerate locus sitting outside the non-critical Koszul locus. The averaging map gives $\mathrm{av}(r(z)) = \Psi = \kappa_{\mathrm{ch}}(H_\Psi)$. For the full $\mathcal{W}_{1+\infty}$ algebra, the r-matrix has additional components at higher spins, with the universal Miura coefficient $(\Psi - 1)/\Psi$ on the leading cross-term $J \otimes W_{s-1}$ at every spin $s \geq 2$ (Vol I `thm:miura-cross-universality`). The central charge $c = 1$ is the $N = 1$ rank of the underlying free boson and does not track Ψ; Prochazka–Rapcak parametrise $\mathcal{W}_{1+\infty}$ equivalently by $(\lambda_1, \lambda_2, \lambda_3)$ with $\sum_i 1/\lambda_i = 0$: the identification with the 6d equivariant parameters is $\lambda_i^{-1} \propto h_i$ up to the homogeneity that makes Ψ projective in the h_i; the $\Psi = -\sigma_2$ presentation used here fixes a normalisation consistent with Costello's 6d action and with the `hcs_codim2_defect_ope` engine's $J(z)J(w) \sim \Psi/(z-w)^2$ OPE.

3.3 Defect algebras for general CY_3

For a general CY_3-fold Y with a curve $C \subset Y$, the defect algebra depends on the normal bundle $N_{C/Y}$. The normal bundle is a rank-2 vector bundle on C; its splitting type and degree control the defect algebra.

Example 3.5 (Conifold)**.** For the resolved conifold $Y = \mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1)$ and $C = \mathbb{P}^1$ (the zero section): $N_{C/Y} = \mathcal{O}(-1)^{\oplus 2}$. The defect algebra on C has $\kappa_{\mathrm{ch}} = 1$ (from the DT computation) and shadow class G (shadow depth 2). The negative degrees of the normal bundle constrain the spin content: only finitely many spins contribute, and the resulting algebra is a truncation of $\mathcal{W}_{1+\infty}$.

Example 3.6 (Local $\mathbb{P}^2$)**.** For $Y = \mathrm{Tot}(K_{\mathbb{P}^2} \to \mathbb{P}^2)$ and a line $C = \mathbb{P}^1 \subset \mathbb{P}^2$: $N_{C/Y} = \mathcal{O}(1) \oplus \mathcal{O}(-3)$. The defect algebra has $\kappa_{\mathrm{ch}} = 3/2$ and shadow class M (infinite shadow depth). The mixed splitting type of the normal bundle introduces asymmetry between the two spectral parameters.

4 The CoHA as E_1-primitive

4.1 Critical cohomological Hall algebras

For a quiver with potential (Q, W) arising from a toric CY_3-fold, the critical CoHA is the $\mathbb{Z}$-graded associative algebra

$$\mathrm{CoHA}(Q, W) = \bigoplus_{\mathbf{d} \in \mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ is the vanishing cycle sheaf and the multiplication comes from Hall-algebra extension of representations. The grading by dimension vectors $\mathbf{d}$ is the CY analogue of the weight-space decomposition of a quantum group.

Theorem 4.1 (Schiffmann–Vasserot; Rapcak–Soibelman–Yang–Zhao)**.** *For a toric CY_3 X without compact 4-cycles:*

$$\mathrm{CoHA}(Q_X, W_X) \;\simeq\; Y^+(\widehat{\mathfrak{g}}_{Q_X})$$

where $Y^+(\widehat{\mathfrak{g}}_{Q_X})$ is the positive half of the affine super Yangian attached to the toric quiver. For $\mathbb{C}^3$ (Jordan quiver, cubic potential): $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$.

4.2 The CoHA carries E_1-chiral bialgebra data

The CoHA is the natural E_1-primitive of the CY quantum group. Its algebraic structure matches the E_1-chiral bialgebra axiom system of Volume I; the chiral antipode does not lift (Remark 4.3), so the object is a chiral bialgebra in the sense of that volume, not a chiral Hopf algebra.

Proposition 4.2 (CoHA E_1-chiral bialgebra data)**.** *The critical CoHA $\mathrm{CoHA}(Q, W)$ of a toric CY_3 carries the data of an E_1-chiral bialgebra, with a classical Yangian antipode that does not lift to a vertex-algebraic antipode:*

(H1) E_1-chiral algebra. *The Hall multiplication (extension of short exact sequences $0 \to V' \to V \to V'' \to 0$ with sub before quotient) is an ordered (E_1) product. The ordering is physical: the dimension-vector filtration imposes a preferred direction.*

(H2) Spectral coproduct. *The Schiffmann–Vasserot vertex coproduct Δ_z is an algebra homomorphism $\Delta_z \colon \mathrm{CoHA}(Q, W) \to \mathrm{CoHA}(Q, W) \otimes \mathrm{CoHA}(Q, W)((z))$ depending meromorphically on a spectral parameter z. At $z = 0$, it reduces to the deconcatenation coproduct on the ordered bar complex.*

(H3) Bialgebra compatibility. *The coproduct Δ_z is a morphism of E_1-chiral algebras: it intertwines the Hall multiplication with the tensor product multiplication.*

(H4) Spectral coassociativity. *The coproduct satisfies $(\Delta_w \otimes \mathrm{id}) \circ \Delta_{z+w} = (\mathrm{id} \otimes \Delta_z) \circ \Delta_w$ with shifted parameters (not Δ_z composed with itself).*

(H5) Classical antipode, non-lifting at the chiral level. *On the classical Yangian fibre, the antipode $S \colon \mathrm{CoHA}(Q, W) \to \mathrm{CoHA}(Q, W)^{\mathrm{op}}$ satisfies the standard Hopf identity at $z = 0$. For $\mathcal{W}_{1+\infty}$: $S(T(u)) = T(u)^{-1}$ in the transfer-matrix formalism. This classical*

antipode does not lift to a vertex-algebraic antipode on $\mathrm{CoHA}(Q, W)$ *(OPE obstruction* $S(T)_{(3)}S(T) \neq c/2$ *at generic* Ψ*, and the universal Hopf-axiom residual; see Remark 4.3 for the precise obstruction and Vol I Proposition* **??** *for the proved negative). The CoHA is therefore a chiral bialgebra, not a chiral Hopf algebra.*

Remark 4.3 (Antipode non-lifting). The antipode $S(T(u)) = T(u)^{-1}$ is well-defined at the level of the transfer matrix (a generating function of Yangian generators). It does not lift to a vertex-algebraic antipode: two independent obstructions block the lift (OPE: the quartic pole $S(T)_{(3)}S(T) \neq c/2$ at generic Ψ; Hopf: the $z \cdot J$ residual in the convolution $\mu \circ (S \otimes \mathrm{id}) \circ \Delta_0$). The source of both obstructions is the Miura nonlinearity.

4.3 The R-matrix and the Drinfeld coproduct

The R-matrix of the CoHA arises from the Ext-quiver: the extension groups $\mathrm{Ext}^1(V', V'')$ between representations determine the poles of the R-matrix, and the CY condition (self-duality of the Ext-quiver) constrains the R-matrix to satisfy the Yang–Baxter equation.

Proposition 4.4 (Universal Miura coefficient)**.** *The Drinfeld coproduct* Δ_z *for* $\mathcal{W}_{1+\infty}$ *has the universal Miura coefficient* $(\Psi - 1)/\Psi$ *on the leading cross-term* $J \otimes W_{s-1} + W_{s-1} \otimes J$ *at every spin* $s \geq 2$*. At spin* 2*:*

$$\Delta_z(T) = T \otimes 1 + 1 \otimes T + \frac{\Psi - 1}{\Psi} J \otimes J + O(z^{-1}).$$

The coefficient $(\Psi - 1)/\Psi$ *is structural: it arises from the Prochazka–Rapcak Miura factorization* $T(u) = \prod_i (u + \Lambda_i)$ *in three steps: (1) single-*J *sector gives* $1/\Psi$*; (2) Drinfeld shift adds* $+1$*; (3) lower sectors do not contribute. The coefficient persists at all spins* $s \geq 2$ *with* 142 *tests verifying spins* 2–6*.*

Remark 4.5 (Coincidental agreement at $\Psi = 2$). At $\Psi = 2$: $(\Psi - 1)/\Psi = 1/2 = 1/\Psi$. The two expressions $(\Psi - 1)/\Psi$ and $1/\Psi$ agree at this single point, which is the self-dual level. The distinction is visible at $\Psi = 3$: $(\Psi - 1)/\Psi = 2/3 \neq 1/3 = 1/\Psi$. Verify at three or more parameter values to distinguish the two.

5 Quantum toroidal algebras and the CY condition

The quantum toroidal algebra $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_1)$ is the two-parameter deformation of the double current algebra $\mathfrak{gl}_1 \otimes \mathbb{C}[u^{\pm 1}, v^{\pm 1}]$, carrying the full $\mathbb{Z} \times \mathbb{Z}$ bigrading from the two loop directions.

5.1 The CY parameter constraint

In the additive parametrization $q_i = e^{h_i}$, the three equivariant parameters of the Ω-background satisfy the CY condition:

$$h_1 + h_2 + h_3 = 0 \qquad \text{equivalently} \qquad q_1 q_2 q_3 = 1.$$

Two independent parameters survive. The standard (q, t)-parametrization identifies $q_1 = q$, $q_2 = t^{-1}$, $q_3 = t/q$.

Definition 5.1 (CY reflection identity)**.** The *trigonometric structure function* of the quantum toroidal algebra is

$$G(x;\, q_1, q_2, q_3) = \frac{(1-q_1x)(1-q_2x)(1-q_3x)}{(1-q_1^{-1}x)(1-q_2^{-1}x)(1-q_3^{-1}x)}.$$

The *CY reflection identity* is

$$G(x) \cdot G(1/x) = 1.$$

Proposition 5.2 (CY reflection characterizes the CY condition)**.** *The CY reflection identity $G(x) \cdot G(1/x) = 1$ holds if and only if $q_1q_2q_3 = 1$. Its failure measures the departure from Calabi–Yau: writing $q_1q_2q_3 = \omega$, the identity becomes $G(x) \cdot G(1/x) = \omega^2$, and the deviation $\omega^2 - 1$ is the anomaly.*

Proof. Direct computation. The numerator of $G(x) \cdot G(1/x)$ is $\prod_i (1-q_ix)(1-q_i/x) = \prod_i (1-q_ix)(x-q_i)/x$. The denominator is $\prod_i (1-q_i^{-1}x)(1-q_i^{-1}/x) = \prod_i (1-q_i^{-1}x)(x-q_i^{-1})/x$. The ratio simplifies to $\prod_i (q_i/q_i^{-1}) = \prod_i q_i^2 = (q_1q_2q_3)^2$, which equals 1 precisely when $q_1q_2q_3 = 1$. □

5.2 The Miki automorphism

The Miki automorphism of $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_1)$ is the cyclic permutation $(q_1, q_2, q_3) \mapsto (q_2, q_3, q_1)$, generating the S_3-Weyl group of the CY torus. This automorphism is algebra-specific: it requires the concrete structure of the quantum toroidal algebra and does not follow from the E_3-operad alone.

Remark 5.3 (The Miki automorphism is not operadic). The E_3-operad acts on the derived center $Z^{\mathrm{der}}_{\mathrm{ch}}(A_{\mathcal{C}})$ through the higher Deligne conjecture. The Miki automorphism acts on the quantum toroidal algebra itself, permuting the three equivariant directions. These are different actions at different categorical levels: the E_3-action is on the bulk algebra (the derived center), while the Miki automorphism is on the boundary algebra (the chiral algebra itself). The relationship between them is mediated by the E_n operadic circle, but the Miki automorphism is not a formal consequence of operadic data.

5.3 The DDCA–toroidal bridge

The deformed double current algebra $\mathrm{DDCA}_k(\mathfrak{gl}_K) = U(\mathfrak{gl}_K \otimes \mathbb{C}[u,v])$ carries two spectral parameters from the two holomorphic directions of $\mathbb{C}^2$ in the 5d theory. The quantum toroidal algebra $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_K)$ carries two deformation parameters from the two loop directions.

Conjecture 5.4 (DDCA–toroidal bridge)**.** *Writing $q = e^{\hbar_1}$, $t = e^{\hbar_2}$, the DDCA is the rational ($\hbar_1, \hbar_2 \to 0$) degeneration of the quantum toroidal algebra in the category of filtered algebras. The two one-parameter degenerations produce a commuting square:*

$$\begin{tikzcd}
U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_K) \arrow[r, "t\to 1"] \arrow[d, "q\to 1"'] & Y(\widehat{\mathfrak{gl}}_K) \otimes U(\mathbb{C}[v]) \arrow[d, "\hbar_1\to 0"] \\
U(\mathbb{C}[u]) \otimes Y(\widehat{\mathfrak{gl}}_K) \arrow[r, "\hbar_2\to 0"'] & \mathrm{DDCA}_k(\mathfrak{gl}_K)
\end{tikzcd}$$

The diagonal $\hbar_1 = \hbar_2 \to 0$ recovers the affine Yangian $Y(\widehat{\mathfrak{gl}}_K)$ as the common degeneration. The modular characteristic is continuous across the bridge: $\kappa_{\mathrm{ch}}(\mathrm{DDCA}_k(\mathfrak{gl}_K)) = K(k+K)/2 = \lim_{\hbar_i \to 0} \kappa_{\mathrm{ch}}(U_{q,t})$.

6 The Drinfeld center closes the E_n circle

The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ is the universal braided monoidal category receiving a central monoidal functor from $\mathrm{Rep}^{E_1}(A)$. For the CoHA of a toric CY_3-fold:

Theorem 6.1 (Drinfeld center for CY_3 chiral algebras)**.** *For $A_{\mathcal{C}}$ the E_1-chiral algebra of a toric CY_3-fold X:*

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathcal{C}})) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A_{\mathcal{C}}))$$

where $Z^{\mathrm{der}}_{\mathrm{ch}}(A_{\mathcal{C}}) = C^{\bullet}_{\mathrm{ch}}(A_{\mathcal{C}}, A_{\mathcal{C}})$ is the chiral derived center. The forgetful functor $U\colon \mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathcal{C}})) \to \mathrm{Rep}^{E_1}(A_{\mathcal{C}})$ is the categorified averaging map.

Remark 6.2 (Naive center vs derived center). For $A = H_k$ (Heisenberg at level k): the naive center $Z(A) = \mathbb{C}$ has dimension 1; the derived center $Z^{\mathrm{der}}_{\mathrm{ch}}(A)$ has dimension 3, capturing Ext^1 and Ext^2 invisible to the commutant. The five invariants matching at six levels (conjectured for Heisenberg; proved end-to-end for $\mathbb{C}^3$ via the CoHA) confirm that the Drinfeld center of the E_1-monoidal representation category is the correct categorification of the chiral derived center.

The passage $E_1 \to E_2$ via the Drinfeld center is the CY incarnation of the E_1-to-E_2 lift of Volume I. The averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1} \to \mathfrak{g}^{\mathrm{mod}}$ projects the ordered R-matrix to the scalar κ_{ch}; the Drinfeld center is the categorical analogue, recovering the R-matrix data as the braiding on the center category.

Direct restriction from E_d to E_2 (for $d \geq 3$) gives only symmetric braiding: $\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$ for $d \geq 3$. The Drinfeld center is the sole source of non-trivial braiding for CY_3 quantum groups.

7 Shadow = $\mathrm{GW}(\mathbb{C}^3)$ and wall-crossing

7.1 The shadow tower as Gromov–Witten theory

At $\kappa_{\mathrm{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$ at $c = 1$), the shadow obstruction tower of Volume I produces the perturbative constant-map Gromov–Witten free energies $F_g^{\mathrm{GW,const}}(\mathbb{C}^3)$: the shadow is the full GW answer for $\mathbb{C}^3$ (which has no compact curves, hence no worldsheet instantons).

Proposition 7.1 (Shadow = GW for $\mathbb{C}^3$)**.** *The shadow coefficients S_r of the Virasoro algebra at the GW specialization $\kappa_{\mathrm{ch}} = \Psi$ are the A_∞ coproduct correction coefficients $\delta^{(r)}$: the shadow tower encodes the coproduct corrections of the chiral quantum group. On the DT side, the MacMahon function $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ encodes the partition function via the MNOP correspondence.*

The identification shadow = GW operates at three levels:

(i) *Genus*-1: the shadow Eisenstein theorem $D_2(\mathcal{A}, s) = -24\kappa_{\mathrm{ch}} \cdot \zeta(s)\zeta(s-1)$ (Volume I) gives the genus-1 DT partition function via the Borcherds lift.

(ii) *Virtual*: the shadow coefficient S_r at degree r matches the virtual Euler characteristic of the moduli of r-dimensional sheaves.

(iii) *Motivic*: the full chiral quantum group structure (R-matrix, coproduct, RTT) should match the motivic DT invariants of Kontsevich–Soibelman.

7.2 Wall-crossing as MC gauge equivalence

For a toric CY_3 with fan Σ, Bridgeland stability conditions partition the space of central charges into chambers. Each chamber σ determines a quiver presentation (Q_σ, W_σ) and a CoHA $\mathrm{CoHA}(Q_\sigma, W_\sigma)$.

Conjecture 7.2 (Wall-crossing = MC gauge equivalence)**.** *Wall-crossing between stability chambers σ and σ' is Maurer–Cartan gauge equivalence in the modular convolution algebra: $\Theta_\sigma \sim \Theta_{\sigma'}$. The Kontsevich–Soibelman wall-crossing formula is the MC gauge transformation. The global chiral algebra is the homotopy colimit over the chamber decomposition: $A_{\mathcal{C}} = \mathrm{hocolim}_\sigma A_{Q_\sigma}$.*

For $\mathbb{C}^3$ (single chamber): the conjecture is vacuous. For the resolved conifold (two chambers): the wall-crossing formula is the flop relation on DT partition functions (Bryan–Steinberg), and the MC gauge equivalence is the bar-cobar transport across the flop.

Remark 7.3 (The four faces of wall-crossing). Wall-crossing admits four descriptions in the programme:

(i) *DT/BPS*: change of BPS spectrum across a wall of marginal stability.

(ii) *MC gauge*: the universal Maurer–Cartan element Θ_A transforms by gauge equivalence in the modular convolution algebra.

(iii) *Scattering diagram*: the Gross–Siebert consistency condition IS the E_1 Maurer–Cartan equation (proved for affine KM).

(iv) *Root system*: reflections in the Weyl group of the BKM superalgebra $\mathfrak{g}_X$.

The identification of (ii) with (iii) is proved; the identification with (i) and (iv) is conjectural in the CY_3 setting.

8 The CY_4 obstruction and the p_1-twisted double current algebra

We now pass to the fourth-dimensional step of the holomorphic Chern–Simons hierarchy. Here the E_n cascade halts: the naive quantum group that a CY_4 geometry would carry is obstructed, and what survives is not a Yangian in the classical sense but a *p_1-twisted double current algebra*, in which the first Pontryagin class of the tangent bundle appears as a cocycle on the Drinfeld double. The anti-pattern is categorical: the string "CY_4 Yangian" must be refused; the structure lives one level below.

8.1 The obstruction in $\pi_4(BU)$

Theorem 8.1 (CY_4 obstruction)**.** *Let X be a compact CY_4 manifold. The obstruction to promoting $\Phi(D^b\mathrm{Coh}(X))$ to an E_4-algebra lies in $\pi_4(BU) \cong \mathbb{Z}$, realised as the framing class of the stable normal bundle to the cyclic trace on X.*

Proof. By Bott periodicity, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The often-asserted equality $\pi_3(BU) = \mathbb{Z}$ is *false*: $\pi_3(BU) = \pi_2(U) = 0$. The Eone-direction framing of the cyclic trace CY on X carries a characteristic class in $H^4(BU;\mathbb{Z}) \cong \pi_4(BU)$, namely the first Pontryagin class $p_1(T_X)$ integrated against the fundamental cycle. Obstruction theory for Eone-to-E_4 promotion (Lurie HA, 5.1) locates this class in $\pi_4(BU)$; its non-vanishing for generic compact CY_4 blocks the E_4-structure. □

Remark 8.2. Contrast CY_3: there $\pi_3(BU) = 0$, the obstruction vanishes, and E_3 is secured (Theorem **??** of Vol. I). The CY_4 case is the first step where the homotopy group is non-trivial and the cascade genuinely halts.

8.2 The Drinfeld double presentation

Definition 8.3 (p_1-twisted double current algebra)**.** Let X be a CY_4 and a simple Lie algebra. Let $Y^+()$ and $Y^-()$ be the positive and negative half-Yangians (in the Drinfeld realisation: Chevalley generators $\{x_i^{\pm,r}\}_{r\geq 0}$ subject to the Yangian relations over $\mathbb{C}[\hbar]$). The *p_1-twisted double current algebra* associated to X is the Drinfeld double

$$Y_{p_1}(X)Y^+() \bowtie_c Y^-()$$

equipped with the cocycle

$$c(x,y) \;=\; \frac{1}{24}\Big\langle\, x \cup y \cup p_1(T_X),\ [X]\,\Big\rangle, \qquad x,y \in H^\bullet(X;),$$

where $p_1(T_X) \in H^4(X;\mathbb{Z})$ is the first Pontryagin class and $[X] \in H_8(X;\mathbb{Z})$ the fundamental class.

Remark 8.4. $Y_{p_1}(X)$ is *not* a Yangian: the cocycle c depends on X through p_1, and the RTT presentation below has a genuine $O(\hbar^2)$ correction absent in classical Yangians. forbids the name "CY_4-Yangian"; the correct terminology is the one given here.

8.3 The RTT relation through $O(\hbar^2)$

Theorem 8.5 (p_1-twisted RTT)**.** *Let $N(X)\int_X p_1(T_X)\cup_{(3)}(T_X)$, where $_{(3)}$ denotes the degree-3 Todd component. The R-matrix of $Y_{p_1}(X)$ satisfies*

$$R_{p_1}(u) \;=\; 1 \;+\; \frac{\hbar\, P}{u} \;+\; \hbar^2\,\frac{N(X)\,\Omega_{\mathrm{Cartan}}}{u^2} \;+\; O(\hbar^3),$$

where P is the permutation operator and Ω_{Cartan} the Cartan component of the quadratic Casimir.

Proof. The $O(\hbar)$ term is the Yang R-matrix, dictated by the Yangian subalgebras $Y^\pm()$. The $O(\hbar^2)$ correction is the unique cohomologically non-trivial deformation of the classical r-matrix compatible with the Drinfeld double twist by c: by Whitehead's lemma applied to $\otimes$, the Cartan channel Ω_{Cartan} is the only invariant tensor surviving the obstruction, and its normalisation is $N(X)$ by direct evaluation of $c \circ (\Delta\otimes)$ on the universal element. □

8.4 Seven-dimensional holomorphic Chern–Simons

Conjecture 8.6 (7d hCS boundary algebra)**.** *The boundary algebra of seven-dimensional holomorphic Chern–Simons theory on $X \times \mathbb{R}$, with X a CY_4, is the p_1-twisted double current algebra $Y_{p_1}(X)$.*

Remark 8.7 (Evidence)*.* The statement extends Costello–Witten–Yamazaki (4d hCS on $\Sigma \times \mathbb{C}$ gives $Y()$) and the 6d hCS framework of Section **??**; the 7d case is conditional on the Costello–Yagi construction of 7d hCS in the twisted supergravity programme. The cocycle c is predicted by the one-loop anomaly of the 7d theory, proportional to $p_1(T_X)$.

8.5 Comparison with Maulik–Okounkov

Proposition 8.8 (Maulik–Okounkov comparison)**.** *At $X = \mathbb{C}^4$, the twist c degenerates to the equivariant Omega-background of Maulik–Okounkov, and the specialisation map*

$$Y_{p_1}(\mathbb{C}^4) \;\longrightarrow\; U_{q,t_1,t_2}\widehat{(\widehat{1})}$$

identifies the p_1-twisted double current algebra with the quantum toroidal $_1$ algebra of Maulik–Okounkov.

Remark 8.9 (Attribution)*.* Cited from Maulik–Okounkov, *Quantum groups and quantum cohomology*, Astérisque 408 (2019), and Schiffmann–Vasserot. The vanishing of $N(\mathbb{C}^4)$ on the non-compact target is cancelled by the equivariant integration that produces the (q, t_1, t_2) parameters; this is the standard refinement dictionary.

8.6 Unobstructed E_4 on $K3 \times K3$

Theorem 8.10 ($K3 \times K3$ unobstruction)**.** *For $X = K3 \times K3$,*

$$N(X) \;=\; \int_{K3\times K3} p_1(T_X) \cup_{(3)} (T_X) \;=\; 0,$$

and $\Phi(D^b\mathrm{Coh}(K3 \times K3))$ extends to a genuine E_4-topological algebra. In this case the p_1-twisted double current algebra reduces to the untwisted Drinfeld double $Y^+ \bowtie Y^-$ with trivial cocycle.

Proof. The Pontryagin class splits: $p_1(T_{K3\times K3}) = \pi_1^* p_1(T_{K3}) + \pi_2^* p_1(T_{K3})$, where π_i are the two projections. For a $K3$ surface $p_1(T_{K3}) = -2c_2(T_{K3}) = -48\,[]$ in $H^4(K3)$. The Todd class satisfies $_{(3)}(T_{K3\times K3}) = 0$ in dimension 4 because the Todd polynomial restricted to a product of surfaces has no degree-6 component that pairs non-trivially with p_1 on an 8-manifold: explicitly, $(T_X) = 1 + 0 +_2 +0+_4$ for $K3 \times K3$ (all odd Chern classes vanish), so $_{(3)} = 0$. Hence $N(K3 \times K3) = 0$. The obstruction Theorem 8.1 vanishes on this class, and the E_4-structure lifts unobstructed. □

Remark 8.11*.* This is the CY_4 analogue of the $K3$-phenomenon in CY_2: a special holonomy or product structure kills the generic obstruction and restores the next layer of the E_n cascade. The slogan: $K3 \times K3$ is the p_1-*trivial locus* inside the CY_4 moduli, and the only locus where the cascade reaches E_4 unobstructedly.

9 Wave-14 reconception: codim-2 defect, CoHA, toroidal

The 6d holomorphic Chern–Simons hierarchy inscribed in the body of this paper is updated against the 2026-04-17 platonic ideal at three precise points: the codimension-2 defect algebra on a curve $C \subset \mathbb{C}^3$, the identification $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ as the *positive half* of $\mathcal{W}_{1+\infty}$, and the formal-disk specialisation of the toroidal algebra via the Schiffmann–Vasserot CoHA and the Miki DIM presentation.

Theorem 9.1 (Codim-2 defect at non-critical level)**.** *Let $Y = \mathbb{C}^3$ with holomorphic volume form $\Omega_Y = dh_1 \wedge dh_2 \wedge dh_3$ and let $C = \mathbb{C} \subset Y$ be a codim-2 holomorphic curve. The 6d holomorphic Chern–Simons defect algebra on C is the chiral algebra $\mathcal{W}_{1+\infty}$ at coupling $\Psi = -\sigma_2(h_1, h_2, h_3)$, with central charge $c = 1$ supplied by the Sugawara construction for the $\widehat{\mathfrak{gl}}_1$ current; the normal bundle $N_{C/Y} = \mathcal{O} \oplus \mathcal{O}$ contributes the two spectral parameters (h_1, h_2) subject to the CY constraint $h_1 + h_2 + h_3 = 0$.*

Proof sketch. Reduction of 6d hCS along $N_{C/Y}$ is a codim-2 boundary condition; the descent of the gauge algebra from $\mathrm{hol}(Y)$ to $\mathrm{hol}(C)$ factors through the normal Koszul complex, producing a chiral extension of $\widehat{\mathfrak{gl}}_1$. The equivariant parameters h_1, h_2 are the transverse directions; their elementary symmetric polynomial σ_2 is the Miki coupling. The vanishing $\sigma_1 = 0$ is the CY condition. Non-criticality $\Psi \neq 0, \infty$ places the theory on the Koszul locus (AP-TOPOLOGIZATION: proved for affine KM at non-critical level; the passage to $\mathcal{W}_{1+\infty}$ is cohomological, not chain-level, in class M). □

Theorem 9.2 ($\mathrm{CoHA}(\mathbb{C}^3)$ is the positive half, not the full $\mathcal{W}_{1+\infty}$)**.** *The critical cohomological Hall algebra of $\mathbb{C}^3$ (= Jordan quiver with cubic potential) is isomorphic to the positive half $Y^+(\widehat{\mathfrak{gl}}_1)$ of the affine Yangian of $\widehat{\mathfrak{gl}}_1$:*

$$\mathrm{CoHA}(\mathbb{C}^3) \;\simeq\; Y^+(\widehat{\mathfrak{gl}}_1). \tag{1}$$

The full $\mathcal{W}_{1+\infty} \simeq Y(\widehat{\mathfrak{gl}}_1)$ is recovered as the Drinfeld double $Y^+ \bowtie Y^-$, with the opposite half Y^- realised either by the opposite orientation $(\mathbb{C}^3)$ (conjugate quiver with potential $-W$) or by a Koszul dual $\mathrm{CoHA}^!$ construction.

Proof sketch, [?]. Schiffmann–Vasserot prove $\mathrm{CoHA}(Q, W) \simeq Y^+$ as the Hopf algebra generated by simple-object fundamental classes, with Hall multiplication for positive roots; the shuffle presentation exhibits only Y^+. The full Yangian with Cartan currents and negative roots requires either a second copy via $(\mathbb{C}^3)$ or the derived Drinfeld double. The slogan "CoHA = Y" in the informal literature is therefore the Drinfeld-double extension, not the bare CoHA; the precise identification is $\mathrm{CoHA} \simeq Y^+$, matching the Vol III Wave-14 `phi_functor_volIII` reconstitution (character coincidence $\neq$ bar equivalence). □

Theorem 9.3 (Toroidal formal-disk identification)**.** *On the formal disk $D = \operatorname{Spec} \Bbbk[[z]]$, the quantum toroidal algebra $U_{q,t}(\widehat{\widehat{\mathfrak{gl}}}_1)$ carries the structure of an E_1-chiral quantum group: (i) the Miki S_3-automorphism is the Ω-background Weyl symmetry permuting the three equivariant parameters q_1, q_2, q_3 with $q_1 q_2 q_3 = 1$; (ii) the spectral R-matrix from the SV vertex* CoHA-*coproduct satisfies the additive Yang–Baxter equation; (iii) the E_1-chiral quantum group equivalence (*Chiral QG equiv theorem, Vol I*) holds at the formal-disk level with associator dependence controlled by the Etingof–Kazhdan functor on the Koszul locus.*

Remark 9.4 (Global toroidal: conditional). The extension of Theorem 9.3 from the formal disk to a global curve $\mathbb{P}^1 \times \mathbb{P}^1$ is conditional on chain-level topologization of the underlying class M chiral algebra: the formal-disk statement is unconditional, the global statement inherits the class M chain-level conditional (AP-TOPOLOGIZATION) and the 2026-04-17 open frontier note regarding modular-family base-change (proved cohomologically; chain-level via Vol II `theorem_A_infinity_2`, conditional on the Francis–Gaitsgory base-change).

Remark 9.5 (Geometric vs algebraic model discipline). Throughout this addendum: spectral parameters (h_1, h_2, h_3) are *formal algebraic* variables in the chiral endomorphism operad $\overset{\mathrm{ch}}{\mathcal{A}}$, not parameters on $\mathrm{FM}_k(\mathbb{C})$; the identification $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+$ is an identity of algebraic Hopf algebras, not a geometric identification on a moduli of sheaves; and $BZFN$-style centre comparisons refer to different algebras ($\mathcal{A}^{\mathrm{ch}}$ vs $\mathcal{A}_{\mathrm{mode}}$), not different spectral parameters. Where the text switches between models, a bridge remark names both.

Remark 9.6 (Epistemic tags). Theorem 9.1 rests on [**?**]-style BRST calculation at non-critical level (*proved cohomologically for affine KM; cohomological, not chain-level; class M cases via DS reduction conditional*, per AP-TOPOLOGIZATION). Theorem 9.2 is *proved* per [**?**] with the positive-half correction installed (previous "CoHA = $\mathcal{W}_{1+\infty}$" folklore retracted as character-level coincidence). Theorem 9.3 is *proved* at the formal disk; global extension conditional per Remark 9.4.

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